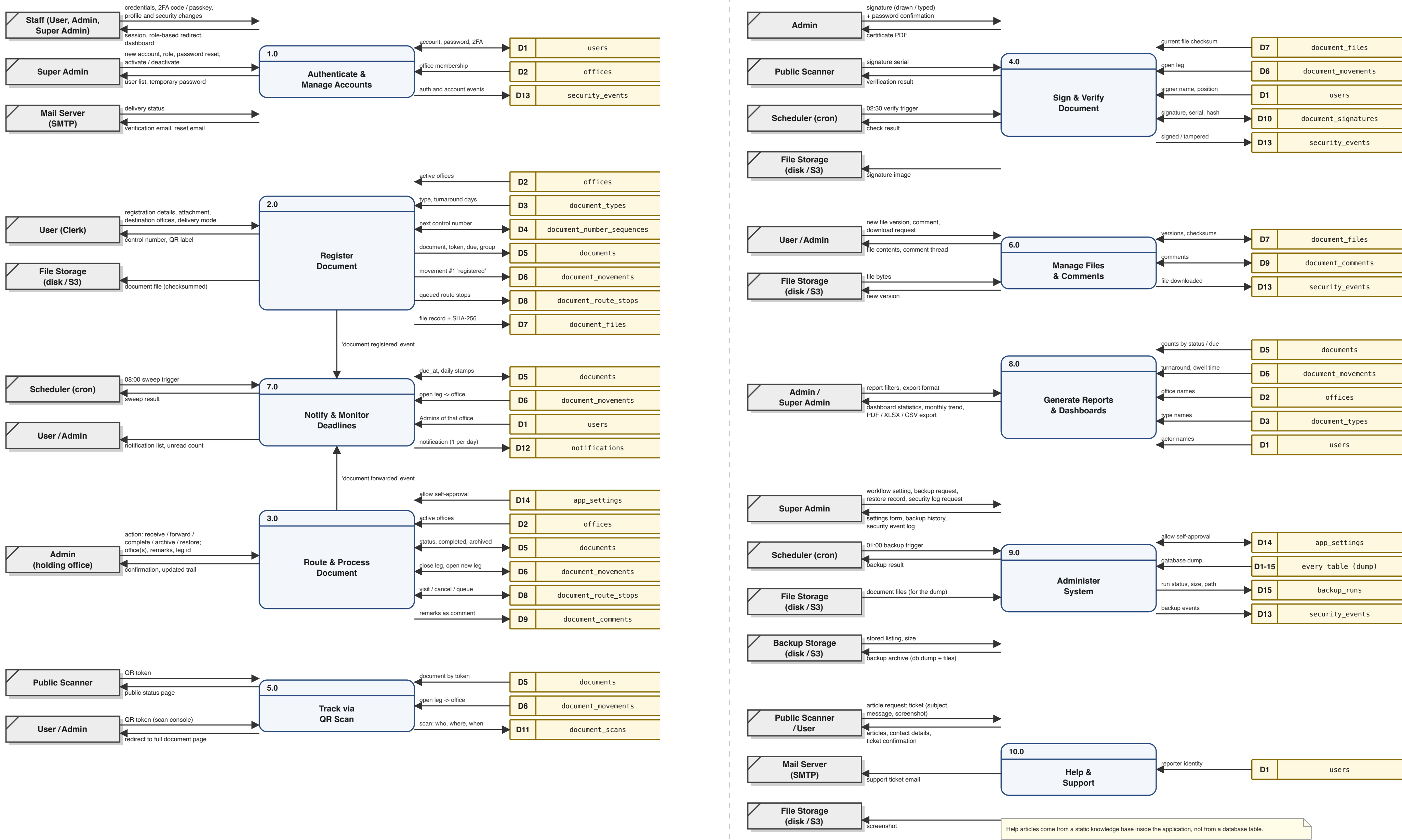


Data Flow Diagram (Level 1)

Process 0 decomposed into ten processes. Notation: Gane & Sarson. Each band shows one process with its external entities (left) and the data stores it reads or writes (right); stores and entities are repeated per band.



Data Flow Diagram (Level 1)

Balancing check: every Level 0 flow and the Level 1 process that carries it.

#	Level 0 flow	Data	Level 1 process
1	User -> System	credentials, 2FA code / passkey	1.0
2	User -> System	registration details + attachment, offices, delivery mode	2.0
3	User -> System	new file version; comment; download request	6.0
4	User -> System	search / filter criteria	8.0 (dashboards and lists)
5	User -> System	QR token (scan console)	5.0
6	User -> System	support ticket (+ screenshot)	10.0
7	User -> System	profile and security changes	1.0
8	System -> User	dashboard; document list, details, trail	8.0 / 3.0
9	System -> User	control number and QR label	2.0
10	System -> User	file download; comment thread	6.0
11	System -> User	in-app notifications	7.0
12	System -> User	help articles; ticket confirmation	10.0
13	Admin -> System	workflow action, office(s), remarks, expected leg id	3.0
14	Admin -> System	signature + password confirmation	4.0
15	Admin -> System	report filters, export format	8.0
16	Admin -> System	office settings	9.0
17	System -> Admin	office queue; action confirmation; updated trail	3.0 / 8.0
18	System -> Admin	certificate PDF	4.0
19	System -> Admin	report export (PDF / XLSX / CSV)	8.0
20	System -> Admin	deadline notifications	7.0
21	Super Admin -> System	new account, role, password reset, activate / deactivate	1.0
22	Super Admin -> System	workflow settings; backup request; restore record; security log request	9.0
23	Super Admin -> System	signature verification request	4.0
24	System -> Super Admin	system-wide dashboard and reports	8.0
25	System -> Super Admin	user list; temporary password	1.0
26	System -> Super Admin	backup history; security event log; settings form	9.0
27	System -> Super Admin	signature verification result	4.0
28	Public Scanner -> System	QR token	5.0
29	Public Scanner -> System	signature serial	4.0
30	System -> Public Scanner	public document status	5.0
31	System -> Public Scanner	verification result	4.0
32	System -> Public Scanner	help / FAQ pages; privacy notice	10.0
33	Scheduler -> System	08:00 deadline sweep trigger	7.0
34	Scheduler -> System	02:30 verify signatures trigger	4.0
35	Scheduler -> System	01:00 backup trigger	9.0
36	System -> Scheduler	job result / log line	4.0 / 7.0 / 9.0
37	System -> Mail Server	verification and reset emails	1.0
38	System -> Mail Server	support ticket email	10.0
39	Mail Server -> System	delivery status	1.0
40	System -> File Storage	document file; signature image; screenshot	2.0 / 4.0 / 6.0 / 10.0
41	File Storage -> System	file contents	6.0 / 9.0
42	System -> Backup Storage	backup archive	9.0
43	Backup Storage -> System	stored listing and size	9.0

Data Flow Diagram (Level 1)

Data dictionary: the fifteen data stores.

ID	Table	Key fields	Read by	Written by
D1	users	id, name, email, password, role (user / admin / super_admin), office_id, position, is_active, two_factor, preferences	1.0, 4.0, 7.0, 8.0, 10.0	1.0
D2	offices	id, code, name, type (office / department / division), parent_id, head_user_id, is_active	1.0, 2.0, 3.0, 8.0	seeded
D3	document_types	id, name, turnaround_days, is_active	2.0, 8.0	seeded
D4	document_number_sequences	office_id, year, last_number	2.0	2.0
D5	documents	id, control_number, qr_token, title, description, remarks, document_type_id, originating_office_id, created_by_id, submission_group_id, status, priority, due_at, completed_at, archived_at, deadline_warned_at, overdue_notified_at	3.0, 5.0, 7.0, 8.0	2.0, 3.0, 7.0
D6	document_movements	id, document_id, sequence, from_office_id, to_office_id, actor_id, action, from_status, to_status, remarks, arrived_at, departed_at, due_at, is_open, ip_address, user_agent	4.0, 5.0, 7.0, 8.0	2.0, 3.0
D7	document_files	id, document_id, version, path, original_name, mime, size, checksum_sha256, uploaded_by_id, movement_id	4.0, 6.0	2.0, 6.0
D8	document_route_stops	id, document_id, position, office_id, status (pending / visited / cancelled), resolved_at, created_by_id	3.0	2.0, 3.0
D9	document_comments	id, document_id, document_movement_id, user_id, context, body, is_internal	6.0	3.0, 6.0
D10	document_signatures	id, serial, document_id, document_movement_id, document_file_id, user_id, signer_name, signer_position, signer_office, method (drawn / typed), purpose, image_path, document_hash, signature_hash, signed_at	4.0	4.0
D11	document_scans	id, document_id, user_id, office_id, source (camera / wedge / manual), ip_address, user_agent, scanned_at	-	5.0
D12	notifications	id, user_id, document_id, document_movement_id, type, dedupe_key, title, body, control_number, read_at	7.0 (read by users)	7.0
D13	security_events	id, type, user_id, subject_id, ip_address, user_agent, context, created_at	9.0	1.0, 4.0, 6.0, 9.0
D14	app_settings	setting_key, setting_value, group_name	3.0, 9.0	9.0
D15	backup_runs	id, status (running / completed / failed), driver, disk, path, size, started_at, finished_at, error, triggered_by_id, restored_at, restore_notes	9.0	9.0